

Vipin Yadav

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EXPERIENCE

Principal Software Engineer

Optym

Jun 2021 – Present

Bengaluru, India

- **Earned two promotions from Software Engineer to Principal Software Engineer**, leading architecture and code reviews across 3 teams, mentoring 10+ engineers, defining engineering standards, and enabling cross-team frontend and backend contributions
- **Created and architected Rune, used by 20+ engineers**, achieving over 90% accuracy in issue and repository understanding, with over 90% of generated PRs merged unchanged after human review, **reducing estimated engineering effort by 80% and direct resolution cost by 75%**
- **Built Rune's multi-repository platform** using OpenHands, AKS, isolated KEDA-scaled workloads, authentication, observability, cost analytics, and synchronized agent instructions across 10+ repositories
- **Architected 3 large-scale React applications central to business-unit revenue**, defining frontend architecture and engineering standards while using Module Federation to reuse a complex graphical application across 3+ products
- **Engineered complex data-intensive product experiences**, including high-volume grids, Gantt and scheduling workflows, scenario management, KPI dashboards, administrative consoles, and role-based access across enterprise railway applications
- **Built a shared MUI component platform adopted across multiple applications**, enforcing reusable design tokens, theming, component standards, and consistent behavior across independently developed product experiences
- **Developed reusable chatbot and batched logging libraries** using Base UI for compatibility across legacy and modern MUI applications, with Biome quality gates, PR validation, versioning, and automated Azure DevOps publishing pipelines
- **Helped recover a multimillion-dollar project after performance-related UAT failure** by scaling WebGL graph visualizations beyond 100,000 interactive data points while preserving smooth zooming and panning
- **Restored real-time workflow responses from minutes to approximately 1–2 seconds** by redesigning React, SignalR, and .NET workflows with request-aware caching, hierarchical notification deduplication, and network-data caching under hundreds of simultaneous refreshes
- **Expanded ownership across 10+ .NET services and APIs**, contributing to service architecture, enterprise integrations, authentication and authorization, observability, production troubleshooting, and end-to-end performance optimization

SKILLS

Languages: TypeScript, Python, JavaScript, C#

AI & Agent Systems: OpenHands, coding agents, agent orchestration, context engineering, LLM routing, Azure AI Foundry, LiteLLM, AI usage analytics, cost governance

Frontend: React, Next.js, TanStack Start, Module Federation, micro-frontends, WebGL, MUI, Base UI, Jotai, MobX-State-Tree, SWR, Jest, React Testing Library, Biome, design systems, performance optimization

Backend & Data: .NET, FastAPI, PostgreSQL, REST APIs, SignalR, asynchronous processing, authentication and authorization

Cloud & Infrastructure: Kubernetes, AKS, KEDA, Azure, Microsoft Entra ID, Azure DevOps, CI/CD, sandboxed workloads, distributed systems, observability

EDUCATION

National Institute of Technology, Sikkim

Bachelor of Technology in Computer Science Engineering

Aug 2017 – June 2021

Sikkim, India